

MIST4610

Group Project

Case#2 – The Live Music Circuit

The Live Music Circuit (LMC) is a company that helps coordinate live shows, tour routing, and equipment logistics for bands performing in and around Athens, Georgia. LMC wants to develop a database to manage venue partnerships, show scheduling, and shared production resources. The LMC operations manager has provided the following information about the current situation:

- LMC works with two main kinds of organizations: venues (places that host performances) and artists (bands or solo performers). Basic information is recorded for all venues and artists.
 - For venues, LMC records: venue ID, venue name, address, phone number, capacity, and whether the venue is all-ages.
 - For artists, LMC records: artist ID, artist name, primary contact phone, and home city.
- Venues may have formal relationships with other venues in the LMC network. These are called venue affiliations (e.g., two venues under the same ownership, venues that regularly co-promote, or venues that share stage gear). A venue can be affiliated with many other venues, and an affiliation involves two venues. Each affiliation has an affiliation type (e.g., *shared ownership*, *co-promotion*, *shared equipment*), a start date, and an optional end date. LMC needs to keep a history of these affiliations, including past affiliations that are no longer active.
- A show event represents a specific night of music at a specific venue (for example, “June 5, 2026 at the Georgia Theatre”). Each show event is defined by: event ID, venue, event date, advertised start time, age policy for that night (which can override the venue’s usual all-ages setting), and whether the event is free or ticketed. A venue cannot have more than one show event with the same advertised start time on the same date.
- Every show event includes one or more artist bookings (e.g., opener + headliner, or a multi-band bill). All bookings (past and present) must be recorded. For each artist booking, LMC records: the artist, the show format, and the artist’s slot order (1 = first opener, 2 = second opener, etc.).
- Each show event is managed by exactly one LMC event coordinator, and has another staff member assigned as backup coordinator.
- LMC also manages production resources that can be allocated to show events. These include items such as PA systems, stage lighting kits, drum risers, microphones, DI boxes, and portable backline amps. Each resource type has a type ID and a name, and it has two pricing rates: one rate for LMC partner venues and another for non-partner venues. (Partner status is tracked per venue.) For example, a “Compact PA System” might be billed at \$75 per event for partner venues and \$125 per event for non-partner venues.
- Each individual resource item (a physical instance) has a unique inventory number and a condition rating (excellent, good, or fair). For example, LMC may own three “Compact PA System” units, each with its own inventory number. Billing is determined by resource type, not by the condition rating.
- All resource allocations (past and present) must be recorded. A show event may require multiple resource items, and the same resource item cannot be allocated to two different show events if their load-in/load-out windows overlap. When resources are assigned to an event, LMC creates a resource allocation record that includes the event, the resource item, the planned load-in time, planned load-out time, and the actual return time (in case the gear comes back late).
- Finally, note that LMC’s work on show scheduling and resource allocations are related but not dependent: a show event can be created and booked without any LMC resources assigned (the venue provides everything), and resources can be allocated for a non-show purpose such as a venue “test night” or community event that LMC supports (these still count as events in the database, but may have no artist bookings).